

EEET202

MACHINES & TRANSFORMERS

MODULE:3

– Types, Losses and Power Flow

of operation of DC motors and select appropriate motor types for

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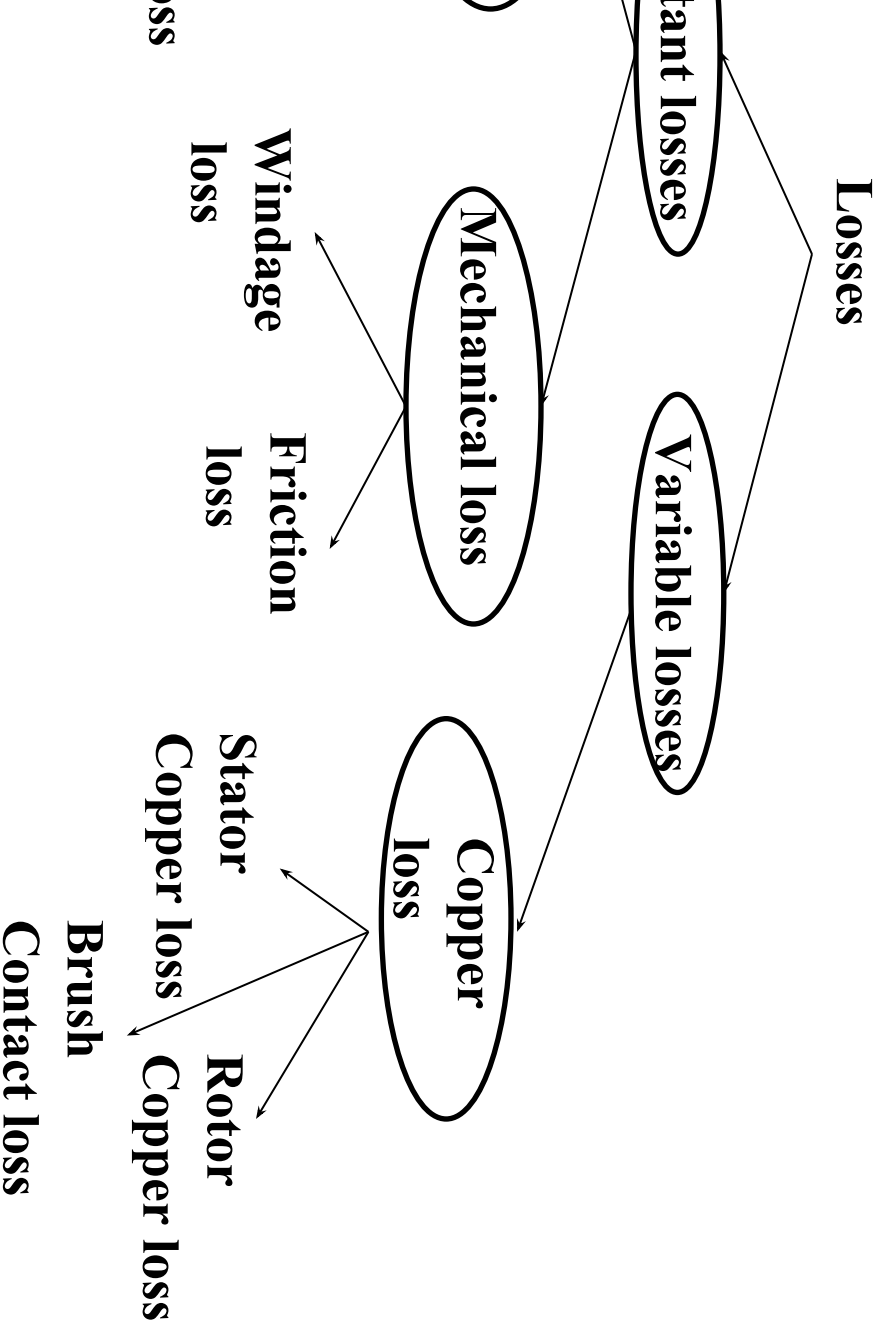
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Losses in DC Motor



Losses in DC Motor

Losses

Motor which remain constant for all loads
constant losses.

Losses

Motor which vary with load are called

Losses in DC Motor

Losses

in the armature core of a DC machine and are of armature in the magnetic field of the poles.

at Loss

Loss

Loss

needed to overcome mechanical friction and wind
drift.

Loss

Losses in DC Motor

Losses

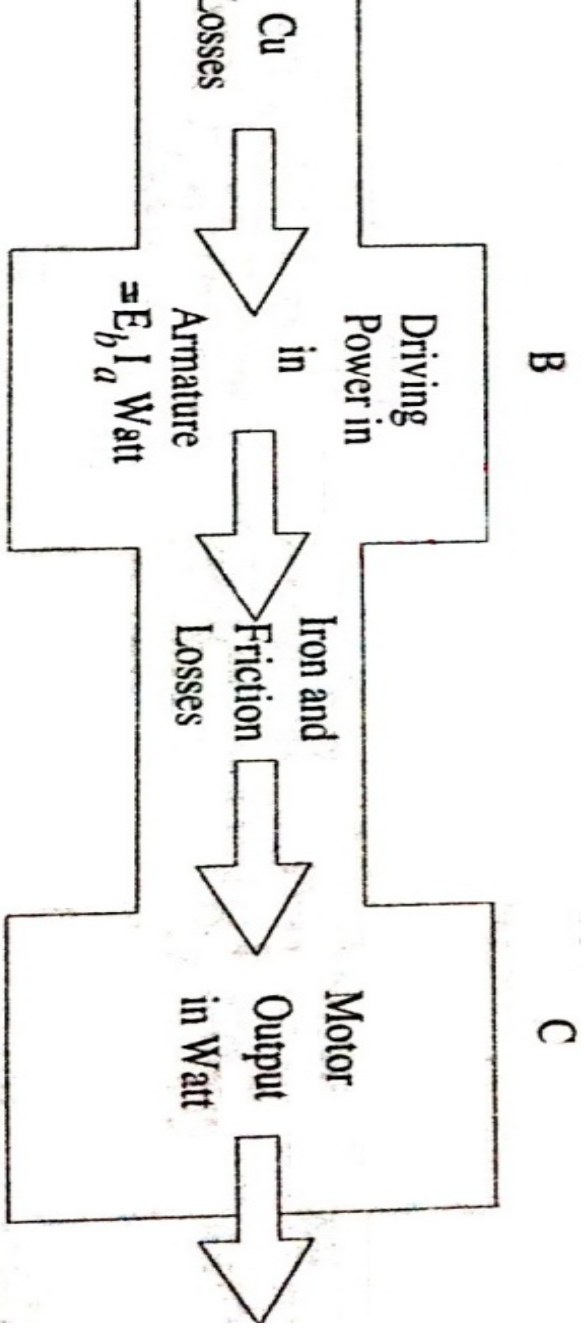
Losses occur due to voltage drop in windings when

$$\text{Copper loss} = I_a^2 R_a$$

$$\text{Copper loss} = I_{sh}^2 R_{sh}$$

$$\text{Copper loss} = I_{se}^2 R_{se}$$

Flow Diagram of DC Motor



is used to determine the efficiency of a generator or motor. From DC Motor, it is clear that the input which is given to the motor is converted into mechanical power in the second stage. The mechanical power.

Efficiencies

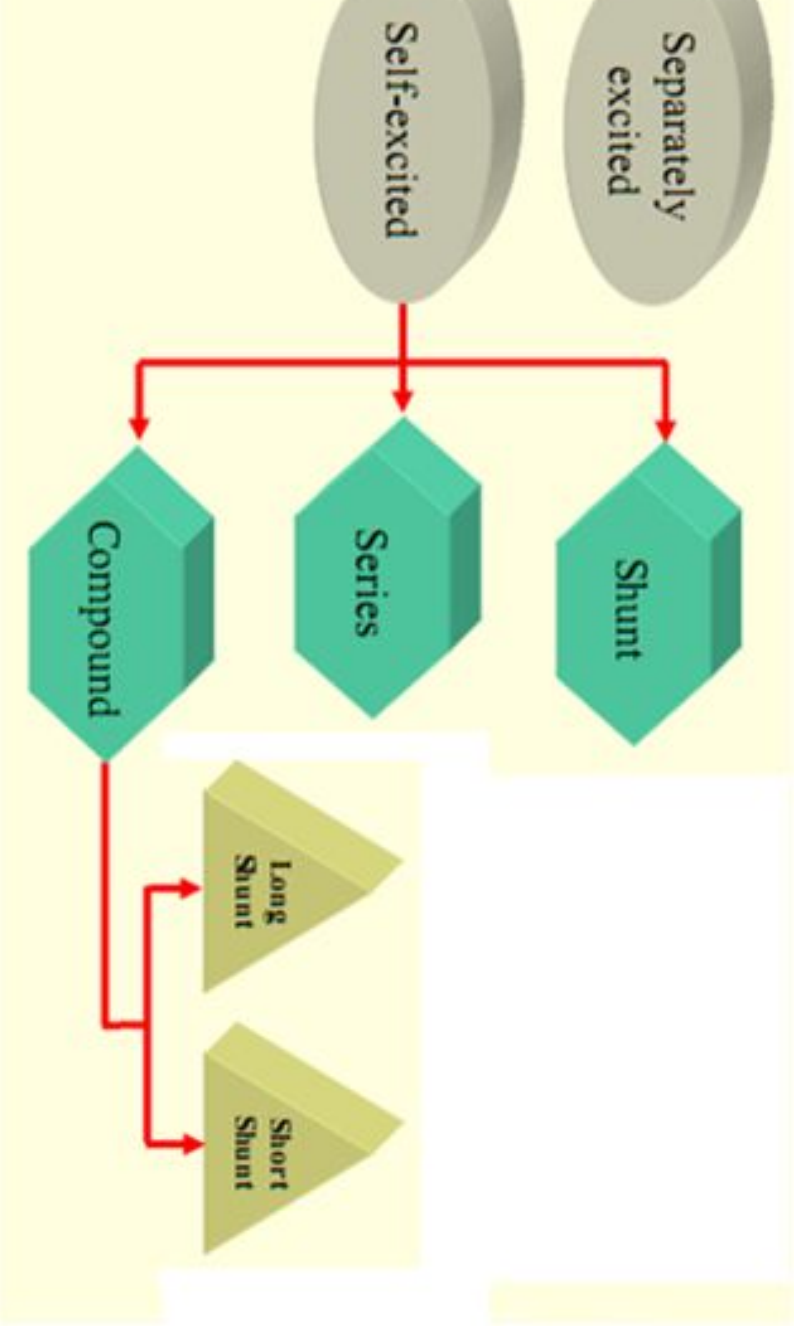
efficiency, η_m = mechanical power
over developed in armature

efficiency, η_e = power developed in
electrical power input

efficiency, $\eta = \eta_m \eta_e$ = mechanical power
electrical power input

Types of DC Motors

of Excitation

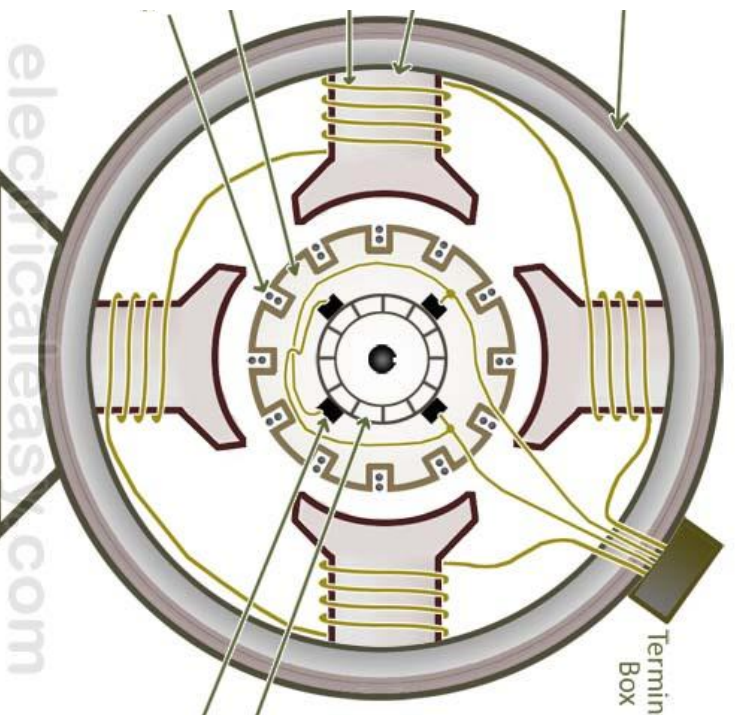


Types of DC Motors

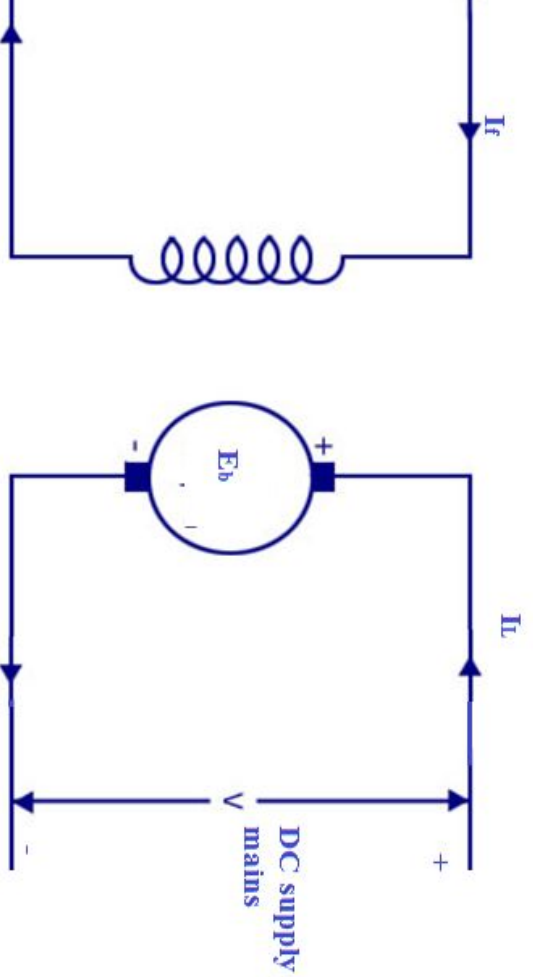
When to take place there should be magnetic field, field winding.

Excited DC Motor – A motor whose field winding is supplied from an independent source other than from the armature source to the motor

Separately Excited DC Motor – A DC motor whose field winding is supplied from a separate DC voltage source of



ately Excited DC Motor



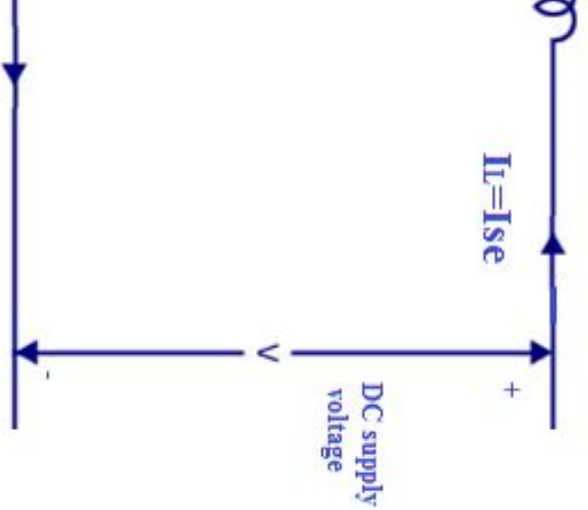
$$I_a = I_L$$

$$V = E_b + I_a R_a$$

Power drawn from supply, $P_{in} = VI_L$

Power developed in armature, $P_a = E_b I_a$

Self Excited DC Motor



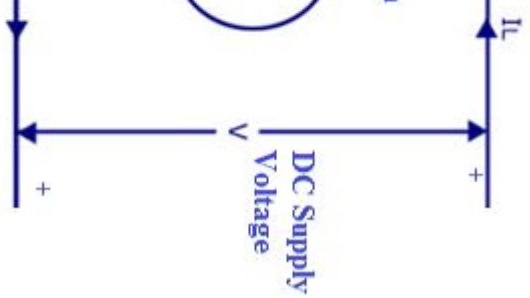
$$I_a = I_{se} = I_L$$

$$V = E_b + I_a(R_a + R_{se})$$

Power drawn from supply, $P_{in} = VI_L$

Power developed in armature, $P_a = E_b I_a$

Self Excited DC Motor



$$I_L = I_a + I_{sh}$$

$$I_{sh} = \frac{V}{R_{sh}}$$

$$V = E_b + I_a R_a$$

Power drawn from supply, $P_{in} = VI_L$

Power developed in armature, $P_a = E_b I_a$

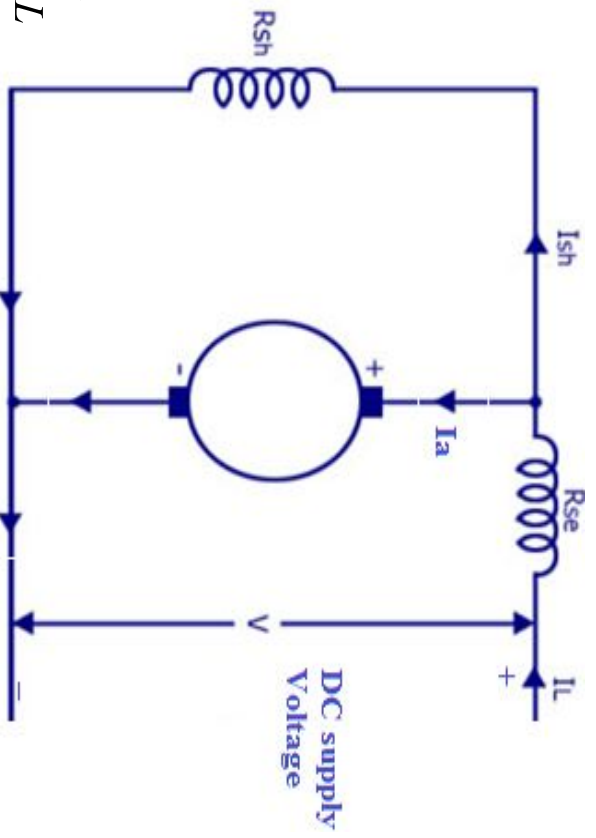
Self Excited DC Motor

Motor

has field winding & shunt field winding

Compound Motor

is parallel with armature windings



R_{se}

supply, $P_{in} = VI_L$

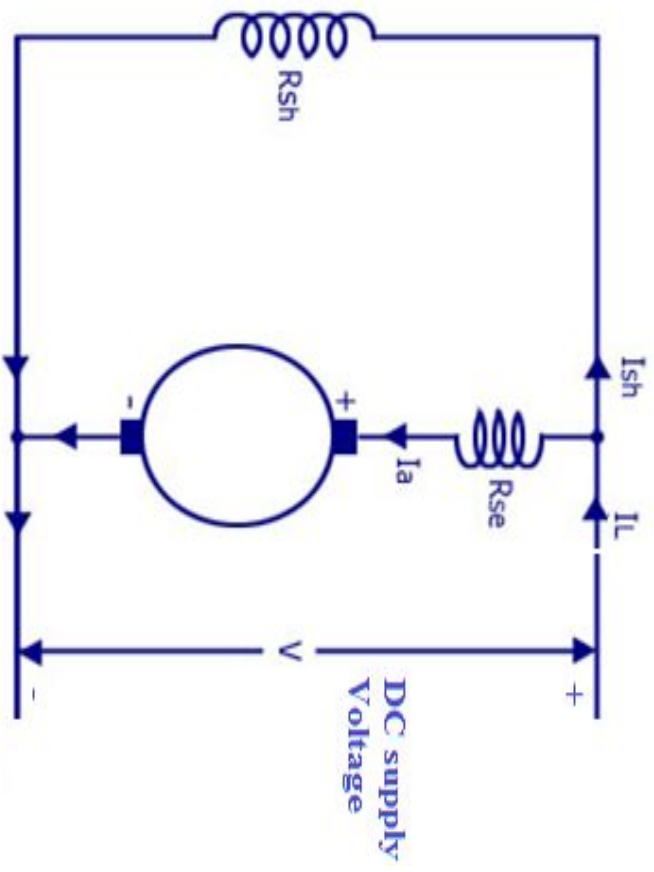
in armature, $P_a = E_b I_a$

Self Excited DC Motor

Motor

Compound Motor

is parallel with both armature and series field



R_{se})

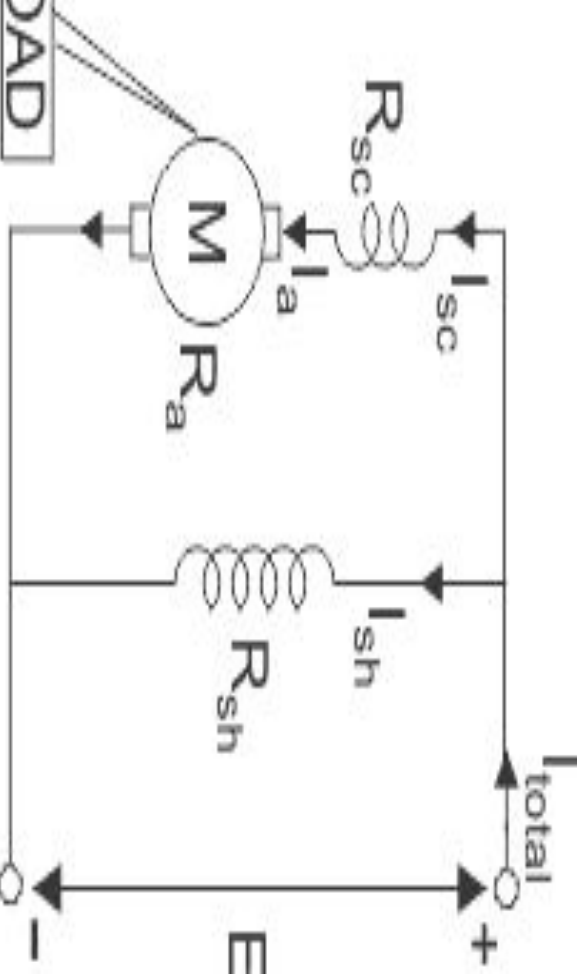
supply, $P_{in} = VI_L$

armature, $P_a = E_b I_a$

Wound DC motor can further be subdivided on the basis of its field winding connection [the armature winding](#), and they are:

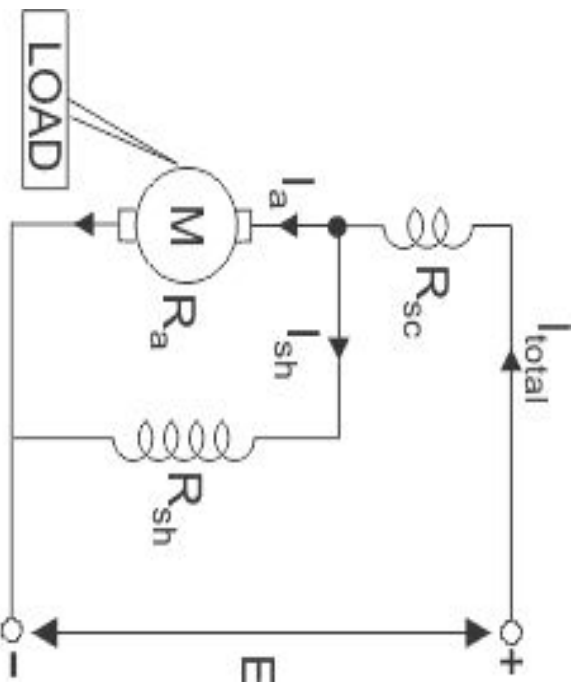
Compound Wound DC Motor

In a compound wound DC motor, the shunt field is connected in parallel across the series field and the armature and series field coil, as shown in the diagram below.



Compound Wound DC Motor

shunt compound wound DC motor, the shunt is connected in parallel across the [armature](#) and series field coil is exposed to the entire before being split up into armature and shunt shown in the diagram below.



Equation of Short Shunt Compound Wound DC

be the total supply voltage and current supplied to the input and I_a , I_{se} , I_{sh} be the values of current flowing through armature winding resistance R_{se} and shunt winding resistance R_{sh} respectively.

Now we can see,

$$I_{total} = I_{se} \dots\dots\dots (2)$$

Current flows through the series field winding.

DC shunt motor,

$$I_{total} = I_a + I_{sh} \dots\dots\dots (3)$$

As the current equation of a short shunt compound wound DC

motor, we apply Kirchhoff's law to the circuit and get,

$$E = E_b + I_a R_a + I_{se} R_{se}$$

$$I_{se} = I_{total}$$

Equation can be written as,

$$E = E_b + I_a R_a + I_{total} R_{se}$$